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PB-12 Strategic BPMN — abstraction level

Backlog item: PB-12
First owner: Ruby
Second owner: Estrella
Date: 2026-09-29
Model: `docs/strategic-bpmn/PB-12_strategic-process_2026-09-29.bpmn`
Operational model this file connects to: `Week3 Coursework/hospital-pathway/bpmn/W02_Hospital_All_Processes_Clean_Lines_Camunda8.bpmn` (`Hospital_All_Processes_Simple_C8`)

> **Current running model (19 October).** The second-release operational model is `Week6 Coursework/bpmn/Hospital_All_Processes_V3.bpmn`, process `Hospital_All_Processes_V3_C8`, and that is what the build packages and deploys. `Simple_C8` above is kept as the name of the model this note was written against on 29 September; the mapping in section 4 is stated in terms of the current model where the two differ, and the differences are marked in that table.

This note explains the abstraction level of the strategic model, how each strategic activity lines up with the executable operational model, and which parts of the full case are left out of the picture.

1. What this model is

The file is a BPMN collaboration for the initial release. The hospital process `Strategic_Hospital_Pathway` has `isExecutable="false"`. It has no Camunda form bindings and no job-worker types. It is not a deployment file. Deploy and demonstrate the operational model

instead: `Hospital_All_Processes_V3_C8` in the second release, `Hospital_All_Processes_Simple_C8` in the initial one.

The picture shows the wider pathway: who receives a request, who is allowed to decide, and which outside organisations the hospital exchanges messages with. Each task is one business stage. The operational model expands that stage into user tasks, gateways, forms and workers.

2. Abstraction rules

1. One strategic task stands for a whole stage. It does not stand for one Camunda user task or one Java worker.
2. A rule that the case states inside a stage is written in that task's BPMN documentation, or on the message flow, instead of becoming another box.
3. The hospital uses three lanes, the same three the operational model uses: Clinical team, Administration, and Finance. Medical secretaries, outpatient bookings, treatment bookings and call handling sit in Administration. The consultant and the clinical nurse specialist sit in the Clinical team.
4. Outside participants are black-box pools. The strategic model does not draw their internal steps.
5. The opening gateway keeps three incoming kinds: referral, enquiry, and change or follow-up. The operational `RequestType` gateway splits those kinds further (for example separate admin, clinical and financial enquiries, and a reports option).
6. After the patient has attended, a parallel gateway starts two results of that attendance: the care plan, and the clinic letter. Payment does not have to finish before the consultant approves the letter. The operational model still runs those steps on one token, with the letter task before the funding task. The strategic split is the business relationship; the operational order is the executable packaging.

3. Pools, lanes and responsibilities

Participant or lane	Responsibility in this model
Referring organisation	Sends referral documents. Receives requests for missing documents, rejection or redirect notices, and a copy of the clinic letter.
Patient or authorised representative	Receives the appointment notice and the approved clinic letter. Receives the reply to an enquiry.
External scheduling service	Offers a slot inside the consultant's window, or reports that no suitable slot exists.
External treatment, laboratory and imaging	Reports whether the requested resource is available. An unavailable resource stays pending.
Insurer or funding organisation	Returns the organisation, approval reference, approved amount and limits.
External payment service provider	Accepts a payment or refund request that does not include the card number, and returns status, reference, date and amount.

Participant or lane	Responsibility in this model
Administration	Checks documents, books and notifies, confirms external treatment resources, distributes the approved letter, and logs enquiries.
Clinical team	Decides the referral, authorises treatment, writes and approves the clinic letter, answers clinical enquiries, and authorises a change or follow-up.
Finance	Determines funding and payment, answers funding enquiries, and reviews a refund or a retained payment.

Clinical staff do not approve refunds. Finance staff do not decide whether treatment is clinically necessary. Call handling answers administrative questions only; clinical questions go to the clinical lane.

4. How strategic activities connect to the operational model

Strategic activity	Operational elements that expand it
Check referral documents	`Register`, `GetDocuments`
Record the clinical decision	`ReviewReferral`, `ReferralDecision`, `Redirect`, `notify-referrer`
Notify the referrer and close	`Redirect`, `EndRejected`
Book the first appointment and notify the patient	`BookVisit`, `VisitNeedsSlot`, `CheckVisitSlot` (`check-slot`), `send-booking-confirmation`. `VisitNeedsSlot` sits between the booking and the slot check: a declined, cancelled or did-not-attend visit skips slot reservation instead of booking a slot it will not use.
Attendance recorded	`AwaitAttendance` holds the booked visit until attendance is known. `VisitOutcome` then splits six ways: attended, pending / retry, patient cancelled, booked awaiting attendance, declined (`RecordDecline`) and did not attend (`RecordDna`). The initial release had one combined cancel/DNA flow; the split is new in the second release.
Authorise the treatment booking	`ClinicalCare`
Determine funding and payment	`Funding`, `FundingOutcome`, `RequestPayment`, `PaymentReceived`, `FundingIssue`, `UrgentAuthorisation`, `mark-payment-investigate`
Confirm external treatment resources	`BookTreatment`, `reserve-appointment`, `flag-resource-unavailable`, `send-booking-confirmation`
Write and approve the clinic letter	Letter fields on `ClinicalCare`, including the revise path back to clinical care
Distribute the approved clinic	`DispatchLetter`, `dispatch-clinic-letter`, `MonitorLetters`

Strategic activity	Operational elements that expand it
letter	
Log the enquiry and route it	`RequestType` enquiry branches, `AdminEnquiry`
Answer the clinical enquiry	`ClinicalEnquiry`
Answer the funding enquiry	`FinanceEnquiry`
Authorise the change or follow-up	`ClinicalChange`, `notify-care-change`
Review the refund or retained payment	`FinanceAdjustment`, `request-refund`

5. What the strategic model collapses on purpose

These rules stay inside a stage. They are not extra boxes.

Rule	Where the strategic model states it
Secretary may request missing documents and must not accept the referral	Check referral documents
Accept, reject, request further information, or redirect, with a reason and a decision-maker	Record the clinical decision. Reject and redirect share the close path.
Letter by default; telephone as well when the visit is within two weeks	Message from the booking task to the patient
No suitable slot stays pending and is not booked outside the clinical window	Message from the scheduling service, and the booking task text
No second booking when an external resource is unavailable	Confirm external treatment resources
Failed or unconfirmed payment is investigated and is not charged again automatically	Determine funding and payment
Urgent care before payment needs a clinical reason and a later finance follow-up	Determine funding and payment. It is not its own branch.
Seven-day letter target; escalation around one month and three months; reminders stop when the letter is sent	Distribute the approved clinic letter. The operational model records the time band on `MonitorLetters` and does not run a multi-day timer.

Rule	Where the strategic model states it
Formal change only; an urgent stop for safety can be authorised afterwards	Authorise the change or follow-up

6. Gaps between the full case, this picture, and the running solution

The strategic model does not draw these, and the running solution does not implement them as enforced behaviour:

Topic	Strategic picture	Running solution
Eight-role login and minimum access	Three lanes only	Every user task is assigned to `demo`. Acceptance test AT-03 fails on purpose.
Immutable audit of every significant action	Not a stage. Audit applies across stages.	Selected workers append `audit_event`. Ordinary users are not refused if they edit the database.
Patient identity matching	Not drawn	`patientId` is free text
Service-outage back-entry	Not drawn	No offline catch-up path
Management reports	Not drawn	`AuditReports` is a human task with a short form
Communication preferences	Not drawn	Forms do not capture postal, digital or accessible-format preferences
Chemotherapy cycle review limited to clinical staff	Not a separate stage. Folded into treatment authorisation.	`ClinicalCare` text and a free-text form; no blood-test fields and no role check
Urgent enquiry highlight	Not drawn	The case has not fixed the urgency rules. The form stores a priority value only.
Full versus partial refund amount	The finance review task does not split the amount	**Closed in the second release.** `finance_adjustment.form` now selects `refund_full`, `refund_partial`, `transfer`, `retain` or `pending`, with `adjustmentAmount` mandatory on every decision except `pending`; `RefundWorkers.requirePartialAmount` rejects a partial amount that is not lower than the original charge. BR-20 is therefore Supported in section 4 of `docs/requirements-traceability-matrix.md`.

External scheduling, treatment resources, funding decisions and card payment are simulated. The strategic pools name those organisations. The operational model draws a pool for the payment provider

and the referrer and the patient; scheduling, laboratory, imaging and the funder appear as workers or form fields rather than as further pools.

7. Second-owner review

Estrella reviewed this note before the item was marked Done.

- ☒ The three lanes can be told apart, and clinical, administrative and financial decisions are not given to the same lane.
- ☒ Referral, consultant decision, first appointment, treatment authorisation, funding and payment, clinic letter, enquiry, and change or follow-up are all on the diagram.
- ☒ Each external message matches a participant in section 3.
- ☒ The mapping in section 4 matches `Hospital_All_Processes_V3_C8`, the current running model.
- ☒ Section 6 still matches the running solution, including AT-03 recorded as Fail.

Second-owner review: accepted — Estrella (Geping Chen), 2026-10-19.
Items 1–3 checked as of 2026-10-19. Items 4 and 5 checked after the corrections A–D.
Running baseline for the mapping: Hospital_All_Processes_V3_C8.
No exceptions left open.
The corrections A–D were applied to the first owner's text under her authorisation, and the first owner (Ruby, Hao Zheng) confirmed the revised text on 2026-10-19. That confirmation was given orally; no separate written note exists.